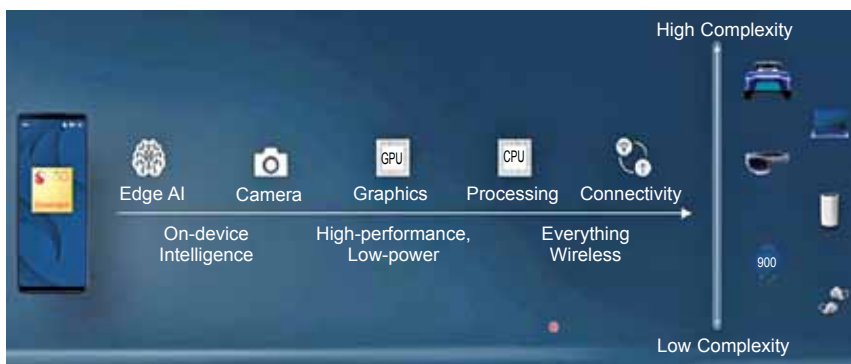




Handsets: One technology roadmap that scales to address all growth vectors (Credit: Ken Wiseman, Qualcomm)

had this strategy that they would reuse technology from their different businesses. So, I was in the wireless business when the people from the CMOS Sensor group came to me and said, “Hey, why don’t we put a camera on the phone?” I thought it was a crazy idea,” he adds. Today, cameras are a huge part of handsets. In terms of the percentage of our silicon, everything revolves around video and camera processing. Without applying Moore’s Law, we would not have the handsets that we have today.

“The end of Moore’s law may be the best thing that has happened in computing since the beginning of Moore’s Law. Confronting the end of an epoch should enable a new era of creativity...”

In terms of mobile technologies, we have modems of various generations of 5G today. We have 802.11, Bluetooth, RF, and many other connectivity technologies. There are so many types of processors—DSPs, GPUs, CPUs, and even neural processors! In multimedia, there is video processing, display, etc. When you think of mobile devices, they have every interaction that you could possibly think of happening on them with a small, compact form factor. And we are constantly adding new features to handsets!

The mobile SoC world: then and now

Previously, in the mobile SoC (silicon-on-chip) world, the focus

was on low-power design and keeping up with the rapid evolution of mobile phone features. Engineers were also focused on the RTL design of IPs. These IPs were largely the CPUs, GPUs, DSPs, and dedicated hardware accelerator IPs that were necessary to save power. Chip architecture was somewhat a newer concept, except in more well-established companies like Intel.

“But overall, Moore’s Law was alive and well with each new tech node. We would get a lot of power and cost savings, and the die would stay the same, or sometimes get a little smaller. Performance would improve significantly with each new tech node,” recalls Wiseman

In the last few years, however, the features of mobile phones have grown exponentially. These features and IPs are more demanding on power and performance. There is artificial intelligence (AI), machine learning (ML) accelerators, heterogeneous compute, and much more. The features are growing, but Moore’s Law is slowing down. Power, performance, and cost improvements are not improving as

We now need creative architectural system solutions to get better power, performance, and cost savings.

fast as they did in the past. Moreover, the die area and package area are increasing due to less scaling.

Today, many phones and their chips have also reached their thermal limits. “If I operate everything on my phone today, all at once, there are going to be difficulties,” Wiseman says. There are a lot of challenges around thermal throttling, and around other design or architecture changes that we have to make to accommodate the thermal capacity of the handset. We are also constrained on bandwidth, since it has increased dramatically with more cameras, 5G, and ML hardware.

Architecture has actually been a big catalyst to solving some of these problems to get better power performance and cost savings. So, we now need creative architectural system solutions to get better power, performance, and cost savings. We can no longer count on the process node to save us. We actually need to work harder with the process nodes to get what we can out of them. And then, we have to work harder on the architecture too, using compression and some software algorithms to manage the system better. We also need to focus on hardware-software trade-offs and try to reduce the silicon footprint. So, that’s today’s mobile SoC world.

Connecting mobile architecture to newer innovations

XR glasses. Augmented reality (AR), virtual reality (VR), mixed reality (MR), and everything in between is all included under extended reality (XR). Although AR and VR offer a variety of ground-breaking experiences, XR is powered by the same underlying technologies as AR and VR. XR glasses have all sorts of features—displays, cameras, sensors, speakers! “The dream here is that XR glasses will have 5G connectivity too, and maybe be able to replace the handsets. It’s a big dream,” Wiseman says.